

PART 5: PROMPT ENGINEERING CHALLENGE

1. Summarising Notes:

Prompt

"You are an expert academic assistant. Read the following notes and create a concise summary. Highlight the key concepts, important definitions, and main points in bullet format. Keep the summary simple and easy to revise for exams."

AI Response

Summary of Artificial Intelligence:

- * Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence.
- * AI applications include speech recognition, image processing, and decision-making.
- * Machine Learning is a subset of AI that allows systems to learn from data.
- * Deep Learning uses neural networks for complex tasks.
- * AI improves efficiency, accuracy, and automation in various industries.

Why the Prompt Worked

- * Clearly defines the AI's role.
- * Specifies the required output format.
- * Focuses on important information only.
- * Produces concise and exam-friendly notes.

2. Generating a Study Plan:

Prompt

"Create a 4-week study plan for learning Python programming. Assume I am a beginner and can study 2 hours per day. Include weekly goals, daily activities, revision sessions, and practice exercises."

AI Response

Week	Topics	Activities
Week 1	Python Basics	Variables, Data Types, Input/Output
Week 2	Control Statements	If-Else, Loops, Practice Programs
Week 3	Functions & Lists	Functions, Lists, Tuples, Dictionaries
Week 4	File Handling & Projects	File Operations, Mini Project, Revision

Daily Study Time: 2 Hours

- * 1 Hour Learning Concepts
- * 45 Minutes Coding Practice
- * 15 Minutes Revision

Why the Prompt Worked

- * Specifies duration and study time.
- * Identifies learner level.
- * Requests a structured schedule.
- * Ensures practical and revision activities are included.

3. Creating Interview Questions:

Prompt

"Act as a technical interviewer. Generate 10 interview questions for a Python Developer position. Include a mix of beginner, intermediate, and advanced questions along with expected answers."

AI Response

1. What is Python?

- * A high-level, interpreted programming language.

2. What is a list in Python?

* An ordered, mutable collection of elements.

3. Difference between list and tuple?

* Lists are mutable; tuples are immutable.

4. What is a function?

* A reusable block of code that performs a task.

5. Explain exception handling.

* It manages runtime errors using try-except blocks.

6. What is OOP?

* Object-Oriented Programming based on classes and objects.

7. What is inheritance?

* A class acquiring properties of another class.

8. What are decorators?

* Functions that modify another function's behavior.

9. Explain multithreading.

* Running multiple threads concurrently.

10. What is a Python module?

* A file containing reusable Python code.

Why the Prompt Worked

* Defines the interviewer role.

* Specifies the job position.

- * Requests varying difficulty levels.
- * Includes answers for preparation.

4. Writing a Resume Summary:

Prompt

"Write a professional resume summary for a Computer Science graduate with skills in Python, Java, SQL, and Machine Learning. Keep it between 80–100 words and suitable for entry-level job applications."

AI Response

Motivated Computer Science graduate with strong knowledge of Python, Java, SQL, and Machine Learning. Skilled in problem-solving, software development, and database management. Experienced in academic projects involving data analysis and AI applications. Quick learner with excellent communication and teamwork abilities. Seeking an entry-level position to apply technical skills, contribute to organizational growth, and gain professional experience in software development and emerging technologies.

Why the Prompt Worked

- * Specifies candidate background.
- * Mentions required skills.
- * Sets word limit.
- * Defines target audience.

5. Creating a Lesson Plan:

Prompt

"Act as an experienced teacher. Create a 60-minute lesson plan on Introduction to Artificial Intelligence for undergraduate students. Include learning objectives, teaching activities, classroom discussions, assessment methods, and learning outcomes."

AI Response

Lesson Topic

Introduction to Artificial Intelligence

Duration:

60 Minutes

Learning Objectives

- * Understand the meaning of AI.
- * Identify AI applications.
- * Explain basic AI concepts.

Activities:

****10 Minutes****

Introduction and discussion on AI.

****20 Minutes****

Presentation on AI concepts and applications.

****15 Minutes****

Group discussion on AI in daily life.

****10 Minutes****

Quiz and question-answer session.

****5 Minutes****

Summary and feedback.

Assessment

- * Classroom participation.
- * Quiz performance.
- * Group discussion contribution.

Learning Outcomes

Students will be able to:

- * Define AI.
- * Identify real-world AI applications.
- * Explain the benefits of AI.

Why the Prompt Worked

- * Assigns a clear teaching role.
- * Defines lesson duration and audience.
- * Requests specific lesson components.
- * Produces a complete and organized lesson plan.